Solving linear equations

Teacher resource

Question 1

$$18 + 9 = a \times b$$

In this equation *a* and *b* are both integers.

How many different values can *b* have to make this equation true?

1	2	4	8
O	O	O	O

Answer: C
Skill: Students solve a simple equation that has more than one solution.

Additional questions

1. How many ways can this equation be solved using only negative integers?
2. Write three ways you could complete this number sentence if *a* and *b* do not need to be whole numbers.

Question 2

$$45 \times m = 18$$

What value of *m* makes this equation true?

$\frac{2}{5}$	$\frac{5}{9}$	$\frac{2}{9}$	$\frac{1}{5}$
O	O	O	O

Answer: A
Skill: Students find a missing number using multiplication or division.
Background information
While students may be able to solve this question by inspection or trial and error, they should be encouraged to consider how the inverse relationship between multiplication and division could be used to solve this problem.

The inverse relationship between multiplication and division is usually more difficult to understand than the inverse relationship between addition and subtraction. Students may understand the inverse relationship for multiplication and division in more obvious cases such as $15 \times m = 45$, but may not be able to generalise this thinking to situations which involve multiplication and division by fractions and decimals.

Additional questions

1. Which number is missing in this equation?
   $n \times 45 = 18$
2. Find the value of p in this equation.
   $18 \div p = 45$

Question 3

What is the value of a in this equation?

$$(124.8 - 24.6) \times a = 803.604$$

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Answer: 8.02
Skill: Students find a missing number in an equation involving decimals

Additional questions

1. What is the value of d in this equation?

   $(124.8 - d) \times 16.04 = 803.604$

2. Find the value of g in this equation.

   $(g - 26.4) \times 16.04 = 803.604$

Question 4

Find the value of n.

$$300 = 800 - \frac{n}{5}$$

60	300	500	2500	5500
O	O	O	O	O

Answer: D
Skill: Students solve an equation with more than one operation

Additional questions

1. What is the value of n in this equation?

   $300 = 800 - 5n$

2. What value of n makes this equation true?

   $800 = 5n - 300$

Student activity: Solving linear equations

Question 1

$$18 + 9 = a \times b$$

In this equation *a* and *b* are both integers.

How many different values can *b* have to make this equation true?

1	2	4	8
O	O	O	O

Question 3

What is the value of *a* in this equation?

$(124.8 - 24.6) \times a = 803.604$

____

Question 2

$$45 \times m = 18$$

What value of *m* makes this equation true?

$\frac{2}{5}$	$\frac{5}{9}$	$\frac{2}{9}$	$\frac{1}{5}$
O	O	O	O

Question 4

Find the value of *n*.

$300 = 800 - \frac{n}{5}$

60	300	500	2500	5500
O	O	O	O	O